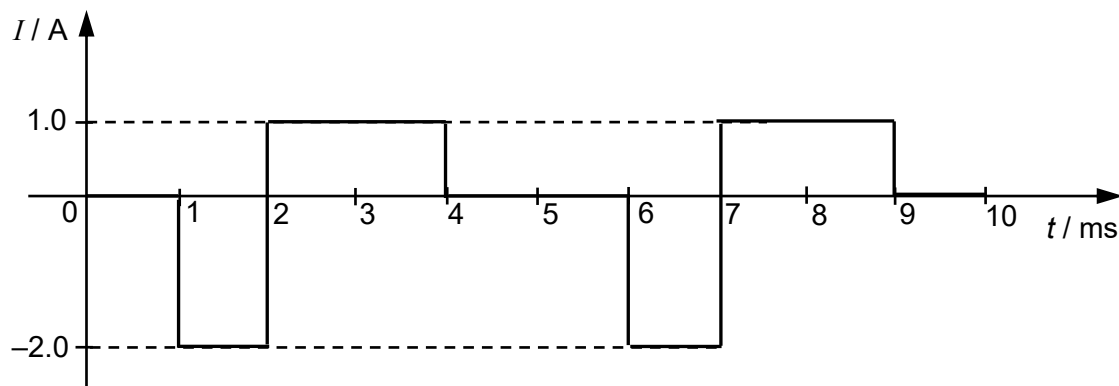


- 26** An alternating current with a rectangular waveform as shown in the diagram below flows through a $10\ \Omega$ resistor.



What is the average power dissipated by the resistor?

- | | | | | | | | |
|----------|-----|----------|-----|----------|------|----------|------|
| A | 0 W | B | 8 W | C | 12 W | D | 28 W |
|----------|-----|----------|-----|----------|------|----------|------|

Answer: C

To find I_{rms} ; Firstly, you square the graph, then you find the total area over 1 period. Next, find the average area over 1 period, and then you will obtain the answer by taking the square root.

$$I_{\text{r.m.s.}} = \sqrt{\frac{(-2.0^2 \times 1) + (1.0^2 \times 2)}{5}}$$

$$= 1.09545 \text{ A}$$

$$\begin{aligned} \text{Average power} &= I_{\text{r.m.s.}}^2 \times R \\ &= 1.09545^2 \times 10 \\ &= 12 \text{ W} \end{aligned}$$